LEONARDO POZZATI

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WHO AM I?

Senior C++ Software Engineer with more than *10 years of experience* specialising in *low-latency*, *performance-critical* systems across *fintech*, *semiconductor*, and *high-tech* industries.
I built proprietary *FX trading infrastructure* with $150 \sim 250\mu s$ *messaging latency* and having a *throughput* of *over 0.8M messages/sec* at *ProMarket*.
Expert in *C++17*, *ZeroMQ*, *lock-free concurrency*, *Linux kernel tuning*, *latency-optim. techniques*.
Proven ability to develop robust software systems using *C*, *C++*, *Python*, and *Bash* in *Agile environments*.
Results-oriented, *collaborative*, and *adaptable team player* experienced in *directly supporting traders* and deploying *live proprietary strategies*, with a *strong academic foundation* in *Mathematical Physics*.

- C / C++
- Python
- Bash
- Linux
- Jenkins
- Git
- OOP/UML
- Agile

EDUCATION

Master's Degree in Mathematical Physics - 110/110 cum Laude — **Universita' Roma Tre**
My research work investigates the *relaxation to the equilibrium* of a *supersaturated gas* through a *critical droplet* which triggers the *phase transition* into liquid.
After writing down both a stochastical and a deterministic *mathematical model* of this metastable thermodynamic system, I wrote the *C++ simulations*, run them and carried out a massive *data analysis in Python* finding out interesting results in the context of *Non-Equilibrium Statistical Mechanics*.

Bachelor's Degree in Physics — **Universita' Roma Tre**

High School — **Pio IX**
Humanities-focused Catholic highschool run by Benedictines Monks.

LANGUAGES

Italian native
English C1 - Academic IELTS
Dutch Self Studying

HOBBIES

I love Art, open air activities like horse riding and hiking, cooking and reading when home.

NON PROFIT

I try to be a better version of me every day, working in charities and being open mostly.

EXPERIENCE

2025 - present — **Software Designer** — **Thermo Fisher Scientific**
- *Developed and maintained automation software* for Transmission Electron Microscopes (TEMs) in the *semiconductor field*, used by Intel and Samsung for *anomaly detection in microchips*.
- Collaborated on *design*, *development*, and *testing* of software in *C/C++*, *Python*, and *Bash*
- Followed *Agile methodologies* throughout the product lifecycle
- Contributed to *optics automation solutions* to improve *system efficiency and robustness*

2024 — **Senior Software Engineer** — **ASML**
- Joined a *core team* responsible for building a *new universal light system* for ASML's lithography machines
- *Designed and implemented* software modules using *C/C++*, with scripting in *Python* and *Bash*
- Followed *Agile practices* to coordinate development, testing, and integration from the ground up

2022 – 2023

Senior Software Engineer **ASML**

- Assigned to the core *Safety and Diagnostic* team for *ASML's High-NA EUV lithography platform*
- Improved system robustness by *integrating*, *refactoring*, and *testing* both *new and legacy code*
- *Collaborated with the T&I* (Test & Integration) team to integrate *CI/CD* pipelines and streamline deployments in the *way of working of the team*
- *Authored the software design documentation* for safety-critical components

2018 – 2021

Senior Software Engineer / Quant Developer **ProMarket**

- Built and owned proprietary *FX trading infrastructure* supporting internal *prop trading teams* actively deploying firm's own capital.
- Designed and implemented a *low-latency routing and messaging infrastructure* using *C++17*, *Python*, and *ZeroMQ* (pub/sub, dealer/router and binary star patterns), achieving stable *latencies of* $150 \sim 250\mu s$ in controlled environments.
- Optimised the messaging infrastructure, reducing end-to-end *median latency by circa* 50% (*from* $400\mu s$ *to* $200\mu s$) via *CPU pinning*, *NUMA-aware memory placement*, *lock-free queues* (`boost::lockfree::queue`), *batched socket writes*, and *lean serialization formats*.
- Scaled message throughput from *around* $400K$ *to over* $0.8M$ *messages/sec* by implementing *NUMA-aware workload partitioning*, *epoll-based event loops (EPOLLONESHOT)*, *custom cache-aligned allocators*, and continuous profiling with *perf*, *Flamegraph*, and *heaptrack*.
- Developed orchestration and coordination services for internal strategy deployment, state management, and system diagnostics (*Python, Bash, Docker*).
- Contributed to a proprietary *sentiment analysis engine* extracting actionable trading signals from financial news and market data, enabling latency and signal quality evaluation in realistic pre-production conditions.
- *Collaborated directly with traders during initial production rollouts*, aligning system performance and operational requirements with trading desk needs.
- Handled *production incident debugging*: resolved market-open latency spike by replacing inefficient `gettimeofday()` *calls with* `clock_gettime(CLOCK_MONOTONIC_RAW)`, *reducing jitter by approximately* 40%.
- Refactored the risk-checking module with a lock-free memory pool, *cutting heap allocation volume by approximately* 30%, stabilising peak memory usage under load.
- Supported live production for more than 18 months, *managing 10–15K orders/day across 3–5 FX trading venues*, deploying multiple strategies in parallel internally.

2015 – 2018

Software Developer **Eveilidia**

- *Maintained and debugged* the backend of custom *on-demand book publishing platforms* deployed on *Linux Fedora*
- Developed and improved *server-side routines* in *Bash* and *Python* for web-based publishing workflows
- Ensured *operational stability* and *performance* of *internal services* supporting client-facing systems

2012 – 2013
freelance

Developer and Tester **Ubuntu Italian Community Forum**

- *Initiated and co-developed* a *command-line parser in C*, inspired by *Python's argparse*, to support Linux scripting users
- Conducted *testing and documentation* for usability within the *Ubuntu Italian user base*
- Published the code as *myoptParser* on GitHub: Vincenzo1968/myoptParser

2010 – 2012

PHP - MySQL Developer **SirCup**

- *Designed, developed, and maintained MySQL databases* to support *data tracking* for *sport tournament operations*
- Implemented *PHP routines* for *dynamic data handling and web presentation* of *tournament results and schedules*
- Contributed to the *automation and customization* of tournament management on the *company's platform*

Miscellanea

After I won a scholarship, I worked for a year as a *librarian* in Roma 3 University library.
I've been *tutoring students in mathematics and physics* for several years, preparing them for any grade level, especially from high school to univeristy.
I worked for a couple of years at Chiostro del Bramante, a museum in Roma, *keeping watch on* *artworks* during exhibitions.

PERSONAL PROJECTS

WOPR (Personal Quant/ML Trading Project)
Built an *end-to-end Python pipeline for Nasdaq futures data*.
From CME *sessionized ingestion* and *range-bar feature engineering* to *TTH label generation*, *multi-model training* (MLP/LSTM/TCN/Transformer), *calibration*, and *reliability metrics*; improved *reproducibility* and *robustness* by unifying *core labeling logic* and using *checkpoint-driven scaling/metadata* across train/eval/inference.

Chat Bot Training
In order to *train a chat bot on a specific domain*, I am writing a Software Engineering Notebook with a massive list of references.
This project *purpose is mostly educative*.

SOCIALS

github.com/vaeVict1s

in/leonardo-pozzati-4769471bb

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